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Project Requirement Document

Project Title : Mediplant Marketplace

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Introduction :

Medicinal plants have historically played a significant role in traditional healthcare systems throughout Bangladesh and continue to be widely used in rural and urban areas alike [1][3]. Due to the country's rich biodiversity, a large number of medicinal plant species are present, many of which are scattered across forests, homesteads, gardens, and agricultural lands [1][2]. However, the supply chain for medicinal plants in Bangladesh remains largely fragmented, underdeveloped, and inefficient, with challenges including erratic market prices, lack of modern processing and transportation infrastructure, and financial and technological constraints faced by farmers and growers [1][4].

The livelihood of smallholder and marginal farmers growing these plants is often unstable. In many cases, middlemen dominate the value chain, resulting in inflated prices downstream and low compensation for primary producers [5]. This dynamic reduces transparency and creates information asymmetry in the market—farmers often lack timely access to market price information or adequate market intelligence, which limits their bargaining power and economic benefit [6].

At the same time, consumer awareness and demand for natural and herbal healthcare options are rising, driven by health consciousness and trust in traditional remedies [3][7]. Despite this growing interest, the lack of a unified, trustworthy digital platform or regulated marketplace in Bangladesh means that buyers and sellers remain disconnected, and the authenticity, quality, and traceability of plant sources are often uncertain [2][4].

These key issues call for a systemic solution—one that can enhance supply chain transparency, empower farmers with real-time or up-to-date market data, reduce dependence on middlemen, and facilitate direct connections between verified plant growers and consumers or healthcare practitioners. The Medi-Plant Marketplace aims to serve as such a digital ecosystem. It proposes a role-based, secure, and user-friendly platform that enables direct listing of plant sources, transparent pricing, order management, and supply chain traceability, thereby addressing core challenges in the sector.

Moreover, by encouraging cultivation and marketplace integration of rare and high-value medicinal plant species, the system can contribute to both economic upliftment of growers and biodiversity conservation. Improved visibility and awareness generation about authentic plant sources may also help reduce overharvesting from wild sources and promote sustainable cultivation practices [4][5]. This report presents a comprehensive development of the Medi-Plant Marketplace, structured around the problem context, objectives, survey analysis, system design and architecture, requirements, and discussion of societal and conservation impact.

Software Requirement Collection:

Questionnaire:

The survey questionnaire was carefully designed to collect data from both sellers (farmers, gardeners, shop owners) and buyers (herbal doctors, medicine makers, and general users). The purpose of these questions was to understand market demand, selling practices, and the overall

acceptance of herbal remedies in society.

Seller information:

What plants do you sell?(Herbal plants/Indoor plants/Rare plants/All)

- Could you please tell me the price range(taka) for selling? (short text)
- Where do you grow your plants and sell them?(Rooftop garden/Nursery)
- How do you prefer to sell your herbal plants?(website/social media/offline)
- Do you think it's useful to have a trusted website where you can sell your mediplant?(short text)
- How do you professionally encourage customers to prefer herbal plants instead of traditional medicine?(short text)

For Seller <i>As a seller provide your feedback to us so that we can able to improve our service and update our system.</i> What kind of plants you want to sell? * ☐ Herbal plants ☐ Indoor plants ☐ Rare plants ☐ All ☐ Other: _____ Could you please tell me the price range(taka) for selling? * Your answer _____	where do you grow your plants and sell? * ☐ Rooftop garden. ☐ Nursery ☐ Other: _____ How do you prefer to sell your herbal plants? * ☐ Website(e.g e-marketplace , daraz,amazon,etc) ☐ Social media (e.g Facebook,instagram ,etc) ☐ Offline (hand-to-hand) ☐ Other: _____
Do you think it's useful to have an trusted website where you can sell your mediplant? Your answer _____ How do you professionally encourage customers to prefer herbal plants instead of traditional medicine?* Your answer _____	

Buyer information:

- How many members are in your family?(short text)
- How many members of your family regularly take medicines?(short text)
- What do you do for headache relief?(take medicine/take hot or cold drinks/ meditation)
- What is your monthly or annual budget for medicine?(short text)
- Do you prefer using herbal remedies over conventional medicine?(short text)
- What kind of plants do you need or prefer to buy ?(herbal plants/indoor plants)
- Where do you prefer to buy herbal plants or remedies?(website/social media/offline)
- What motivates you to choose herbal remedies over conventional medicine?(short text)
- Do you think it's useful to have a trusted website where you can buy your Medicinal plants which may cure your suffering?(short text)

For Buyer <i>As a buyer provide your feedback to us so that we can able to improve our service and update our system.</i> How many members are in your family? Your answer _____ How many members of your family regularly take medicines? * Your answer _____	What kind of plants do you need or prefer to buy ? * ☐ Herbal plants ☐ Indoor plants ☐ Rare plants ☐ Other: _____ Where do you prefer to buy herbal plants or remedies? * ☐ Website ☐ Social media ☐ Offline ☐ Other: _____
What do you do for headache relief? * ☐ Take medicine. ☐ Take any hot drinks. ☐ Take any cold drinks. ☐ Meditation ☐ Other: _____ What is your monthly or annual budget for medicine? * Your answer _____ Do you prefer using herbal remedies over conventional medicine? * Your answer _____	What motivates you to choose herbal remedies over conventional medicine? * Your answer _____ Do you think it's useful to have an trusted website where you can buy your Medicinal plants which may cure your suffering? Your answer _____

Proposed Solution:

The Medi-Plant Marketplace is a complete and integrated solution addressing the fundamental challenges faced by both medicinal plant growers and buyers in Bangladesh. It offers a unified online platform that connects verified sellers with genuine buyers, ensuring fairness, trust, and transparency in transactions. By removing the reliance on unorganized offline markets and social media, the system reduces the exploitation of both sellers and buyers. One of the core strengths of this platform is its secure user authentication system, which ensures that only verified sellers can list plants. This prevents fraudulent activities and builds confidence among buyers. Sellers are empowered to create detailed listings with plant type, price, and contact information, ensuring buyers have access to accurate and transparent data. At the same time, buyers benefit from a streamlined browsing and ordering process, reducing the time and effort required to find authentic medicinal plants. The system incorporates a role-based access model, meaning that buyers and sellers only interact with the features relevant to them. For example, sellers manage plant listings and order responses, while buyers browse shops, place orders, and track deliveries. This not only enhances usability but also ensures data protection and secure operations. To make the platform more inclusive, additional features such as delivery options, profile management, and a dashboard for both separately with important information are integrated. These features ensure that the marketplace is not just an information hub, but also a functional platform that supports real-world transactions. Moreover, the system is designed with scalability in mind, so more sellers and buyers can join as the demand for herbal remedies grows. The Medi-Plant Marketplace also acts as an awareness and conservation tool, spreading knowledge about the importance of rare medicinal plants. By connecting farmers to wider markets, it provides financial incentives for growers to cultivate and preserve near-extinct species, thus contributing to biodiversity protection. In addition, the platform reduces over-dependence on synthetic medicines by giving people an accessible alternative rooted in natural remedies. In times of crisis (such as the COVID-19 pandemic), the system can support healthcare needs by ensuring the continued availability of herbal plants through online orders, minimizing the impact of supply chain disruptions. By enhancing operational transparency, eliminating middlemen, and empowering both farmers and consumers, the Medi-Plant Marketplace creates a fair, efficient, and socially impactful ecosystem. It not only strengthens the supply chain of medicinal plants but also ensures that traditional knowledge and resources are preserved for future generations.

Software requirement specification:

Functional Requirements:

Stakeholders

- **Seller (Farmer / Nursery Owner / Gardener):** Can register, manage their shop, add/remove plant listings, edit details, and set prices. They can also manage delivery options.
- **Buyer (Individual / Household / Herbal Doctor / Pharmacy):** Can register, browse

plants by type, price, or location, view seller shop details, add products to the cart, and place orders with delivery options.

Registration and Authentication

- Each user (Seller or Buyer) must register with their required details (name, phone, contact info).
- Users must log in with secure authentication to access their dashboards and role-based functionalities.

Homepage

- The homepage is accessible to all, excluding non-registered visitors.
- It displays a list of available medicinal plants, their prices, and seller shops.
- A search and filter option (by plant type) is provided for easy navigation.

Dashboard Functionality (Role-Specific)

Seller Dashboard

- **Shop Management:** Add new plants with name, price, type, and location
- **Plant Listings:** Edit or remove plants from the shop.

Buyer Dashboard

- **Browsing & Search:** Browse plants by type
- **Cart Management:** Add plants to cart or remove items.
- **Order Placement:** Confirm purchase, select delivery option

Non-Functional Requirements

1. **Scalability:** The system must support increasing numbers of sellers, buyers, and plant listings as the platform grows.
2. **Security:** Sensitive user data (contact) must be encrypted and protected with secure authentication standards.
3. **Reliability:** The platform should remain available 24/7, especially during high-demand situations (e.g., health crises).

4. **Usability:** The interface must be intuitive, simple, and mobile-friendly to support non-tech-savvy users, with potential for multilingual support.

Performance: The system should process real-time transactions and updates efficiently to support smooth buyer-seller interactions.

USE CASE DIAGRAM:

The Use Case Diagram represents the main interactions between system actors (Buyer, Seller, and Admin) and the functionalities of the *Medi-Plant Marketplace*. Sellers can register an account, log in/log out, manage their profile, add and remove plant listings (with pictures, prices, and details), cancel orders, and choose delivery options (home delivery or pickup). Buyers can register, log in/log out, manage their profile, browse/search plants (based on type, price, or location), view seller contact information, add plants to the cart, and place orders. Buyers can also select their preferred delivery option. Admin is responsible for higher-level system operations, such as verifying user accounts, ensuring data security, and canceling buyer/seller accounts if necessary.

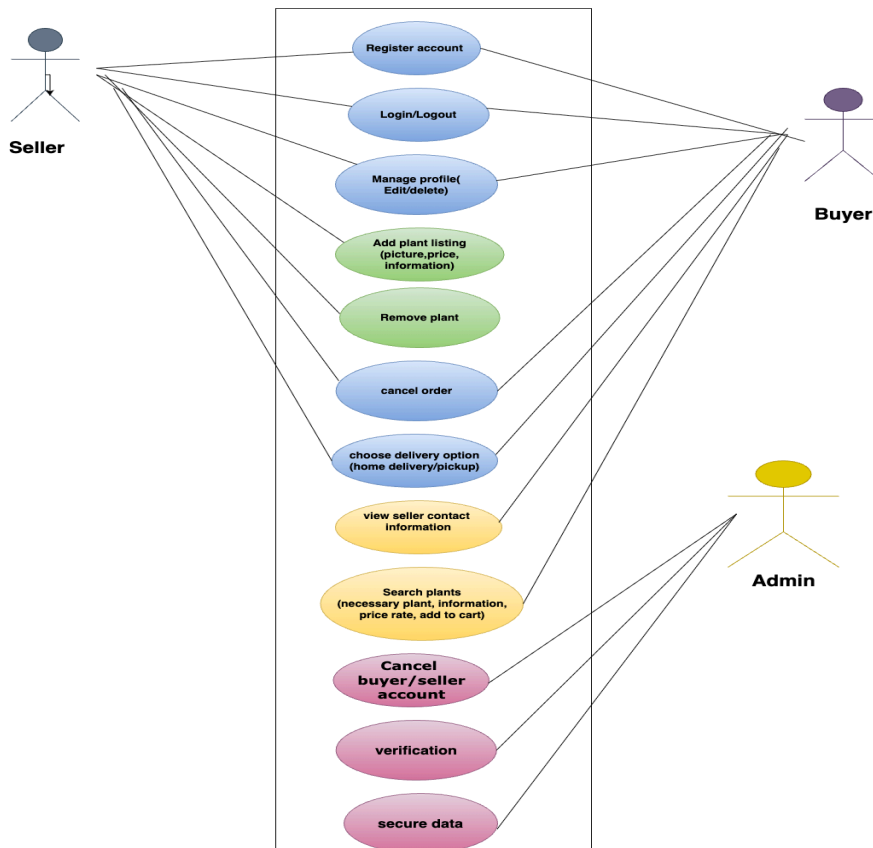


Figure:1 Use Case Diagram-Mediplant market place.

This diagram highlights the role-based access control of the system, ensuring each actor can only perform actions relevant to their role. It also ensures system security and transparency by centralizing verification and data protection responsibilities under the Admin role.

Activity Diagram:

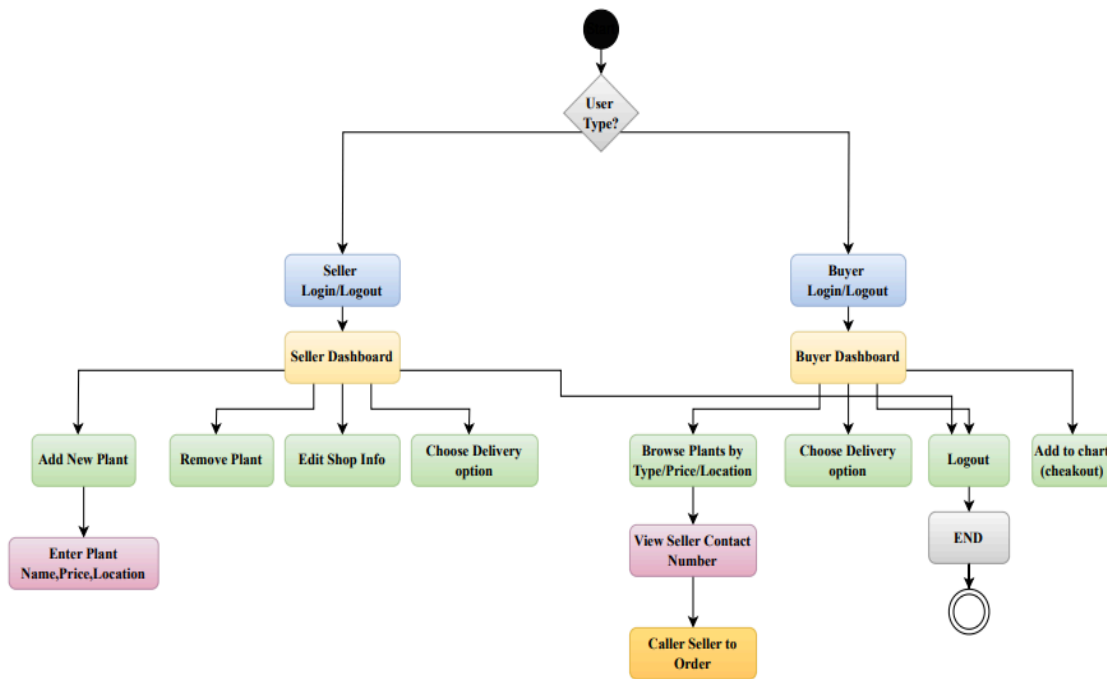


Figure:2 Activity Diagram-Mediplant market place.

The Activity Diagram illustrates the workflow of the system, showing how buyers and sellers interact with the platform from start to end. The system begins with the user type identification (Buyer or Seller). If the user is a Buyer, they can log in, access the Buyer Dashboard, browse plants by type, price, or location, view seller contact details, add items to the cart, and select a delivery option. The buyer can then proceed to checkout and confirm the order. If the user is a Seller, they can log in to the Seller Dashboard, add new plants (with details such as name, price, and location), edit shop information, remove plants, or manage delivery options. The system flow also allows buyers to directly call sellers to confirm orders and finalize delivery. Both types of users can log out when finished, ending the activity sequence. This activity flow ensures smooth interaction between buyers and sellers, while also maintaining clear transaction steps and user accountability.

CLASS DIAGRAM:

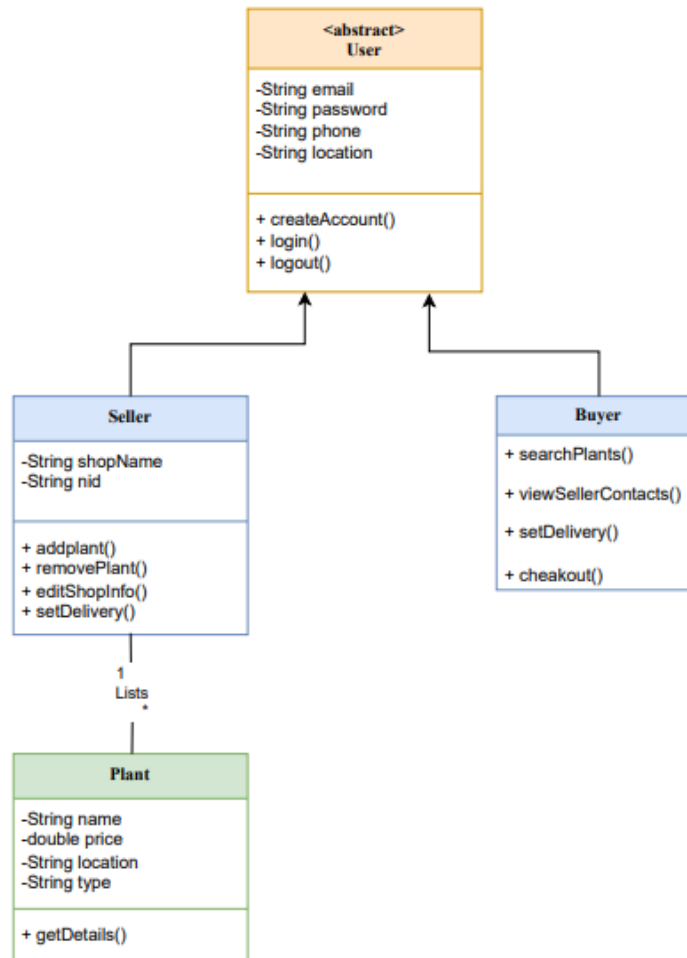


Figure:3 Class Diagram-Mediplant market place.

The class diagram represents a plant marketplace application where users can interact as either buyers or sellers. At the core, there is an abstract **User** class that contains common attributes such as email, password, phone, and location, along with basic functionalities like account creation, login, and logout. Both **Seller** and **Buyer** classes inherit from this **User** class, gaining the shared properties while extending them with role-specific operations. A **Seller** manages a shop by providing details like shop name and national ID, and is able to add or remove plants, edit shop information, and set delivery options. On the other hand, a **Buyer** can search for plants, view seller contact information, set delivery preferences, and proceed to checkout when purchasing items. The system also defines a **Plant** class with attributes such as name, price, location, and

type, along with a method to display its details. Sellers are associated with multiple plants, while buyers interact with both sellers and plants through searching, viewing, and purchasing. This structure provides a clear separation of responsibilities, ensuring that sellers focus on shop and product management while buyers engage in exploring and purchasing plants, all under the unified framework of the User class.

Project management:

Detailed Project Management Plan:

Project Objectives

- Build a scalable, secure, and transparent digital marketplace for medicinal plants.
- Empower sellers (farmers, gardeners, nursery owners) and buyers (households, pharmacies, herbal practitioners) to trade plants efficiently.
- Promote health awareness and biodiversity conservation through accessible herbal plant trading.
- Ensure fair pricing and accountability across the marketplace by eliminating middlemen.

Initiation

- Conduct market research on existing plant trading practices.
- Perform stakeholder analysis to identify the needs of sellers, buyers
- Establish the project scope and objectives.
- Secure resources (technical, human) for project execution.

Planning

- Create a detailed project timeline and define milestones.
- Design use case, activity, and class diagrams.
- Develop a risk management and mitigation strategy.

Execution

- Develop system architecture and role-based dashboards (Seller, Buyer).

- Implement registration, authentication, browsing, cart, and order modules.
- Build a database for storing plant listings, user data, and transactions.
- Ensure mobile-friendly UI for both urban and rural users.

Monitoring and Controlling

- Collect success and failure metrics after each sprint.
- Obtain continuous feedback from test users (buyers and sellers).

Closure

- Demonstrate the completed system to target stakeholders.
- Deliver user guides and technical documentation.
- Collect feedback and prepare a scaling plan for future improvements.

Resource Allocation

- Human Resources: Project manager, software developers, UI/UX designer, database administrator, quality assurance testers, survey coordinators.
- Technical Resources: Development tools (VS Code, Figma, JavaFX, MySQL), servers for hosting, GitHub for version control.
- Other Resources: Survey forms, stakeholder communication tools, marketing platforms.

Timeline

Task	Duration (Weeks)
Requirement Collection	1 Week
System Design (UI + UML)	1 Week
Frontend Development (Buyer)	1 Week
Frontend Development (Seller)	1 Week

Error Handling for Frontend	1 Week
Backend Development	1 Week
Error Handling for Backend	1 Week
Create database file	1 Week
Total	8 week

Gantt Chart :



Risk Management

- Technical Risks: Addressed by regular code reviews, backups, and rigorous testing.
- User Adoption Risks: Reduced by providing user-friendly design, training, and multilingual support.
- Budget Risks: Controlled through contingency funds and scheduled financial reviews.
- Security Risks: Minimized using encrypted authentication and secure data storage.

Deliverables

- Fully functional Medi-Plant Marketplace system (Buyer, Seller).
- UML diagrams (use case, activity, class).

- Training materials and user manuals.
- Post-deployment maintenance and support plan.

Full Project Plan :

Phase 1: Requirement Analysis

- Collect survey feedback from sellers, buyers, and admins.
- Define functional and non-functional requirements based on user needs.

Phase 2: System Design

- Design architecture, workflows, and data models.
- Prepare UI mockups and UML diagrams.

Phase 3: Development

- Build role-based dashboards for Buyer and Seller.
- Implement authentication and secure login.
- Add core modules: Shop Management, Browsing, Cart, Order Placement.

Phase 4: Testing

- Perform unit and integration testing.
- Conduct user acceptance testing (UAT) with selected stakeholders.

Phase 5: Deployment

- Launch pilot version in a small community.
- Monitor early adoption and resolve issues.

Phase 6: Training and Documentation

- Train sellers and buyers on platform usage.

- Provide documentation for system maintenance.

Phase 7: Scaling and Feedback Integration

- Roll out the system to more users across multiple regions.
- Integrate improvements based on user feedback.

Budget & Total cost:

Scenario	total (₹ BDT)
Low (student/freelancer)	₹1,87,900
Mid (small firm)	₹ 3,97,600
High (enterprise)	₹ 7,61,220

For Mid Scenario we assume,

- 2 Java devs → ₹ 2,41,640
- Quality Assurance (part-time) → ₹ 34,950
- PM (part-time) → ₹ 31,455
- Labor subtotal → ₹ 3,316,0
- Total cost → ≈ ₹ 341,205

Contribution to Society:

The Medi-Plant Marketplace goes beyond being a simple online platform — it serves as a socially impactful initiative that benefits multiple groups within society.

1. Empowering Sellers (Farmers, Gardeners, and Nursery Owners):
The platform ensures that not only farmers but also individuals with small rooftop gardens or private nurseries can sell their medicinal plants directly to buyers. This expands income opportunities to a wider population and motivates more people to engage in plant cultivation.
2. Increasing Buyer Accessibility:

Buyers — including households, traditional healers, pharmacies, and health-conscious individuals — gain easy access to authentic medicinal plants that were previously difficult to find. This strengthens trust in natural healthcare practices.

3. Promoting Environmental Sustainability:

By creating a digital ecosystem for selling and buying plants, the system indirectly encourages the preservation of biodiversity. Rare species of medicinal plants receive more value in the marketplace, motivating cultivators to protect and propagate them instead of abandoning them.

4. Reducing Exploitation:

The platform cuts down the role of intermediaries, ensuring fair trade and price transparency. This fosters a sense of justice and trust among both sellers and buyers.

5. Awareness and Education:

The system does not only act as a market but also as an awareness tool. Through the platform, people learn about the benefits of herbal remedies and their importance in maintaining a healthy lifestyle.

6. Digital Inclusion:

With features that can be adapted for mobile notifications or simplified access, the system is designed to support users who may not be highly digitally literate, bridging the digital divide in rural and semi-urban areas.

In this way, the Medi-Plant Marketplace has a holistic contribution to society, reaching beyond economic empowerment to include healthcare benefits, environmental sustainability, and social awareness.

Conclusion:

The Medi-Plant Marketplace demonstrates how technology can be applied to address real-world challenges in healthcare, agriculture, and trade. The platform is not limited to farmers alone — it provides opportunities for any individual with a nursery, rooftop garden, or personal cultivation of medicinal plants to connect with buyers. This inclusivity ensures that both large-scale and small-scale growers have a fair chance in the market.

For buyers, the system simplifies the process of finding authentic medicinal plants. It reduces dependency on unreliable sources and ensures quality, transparency, and trust in every transaction. For sellers, it guarantees visibility, fair pricing, and a wider customer base without needing to rely on middlemen.

From a societal perspective, the project promotes healthier lifestyles by encouraging the use of natural remedies, and it supports biodiversity conservation by creating market value for rare plants. In times of crisis — such as pandemics or disruptions in supply chains — the platform

can play a crucial role in ensuring the continued availability of herbal resources.

Ultimately, the Medi-Plant Marketplace is more than just a commercial application; it is a social innovation that integrates health, economy, and environment into one digital ecosystem. By contributing to fair trade, conservation, healthcare, and economic empowerment, it shows how ICT solutions can build a healthier and more sustainable future for all.

Reference:

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ContributionTable:

Umme Mehajabin Anisha (233014020) : Founder of the MediPlant Marketplace, Build Project Structures, handle backend , create fronted (UI) and error handling .

Tanvin Rima (233014237) : Handle backend , create fronted (UI) and error handling.